PROJECT REPORT

On

"TRANSFORMER BASED RAIL TRACK DEFECT DETECTION"

Submitted For Partial Fulfillment & Award of

BACHELOR OF TECHNOLOGY

In

Artificial Intelligence and Machine Learning

By

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Under the Guidance of

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DEPARTMENT OF COMPUTER SCIENCE (AI & ML) ENGINEERING

RAGHU ENGINEERING COLLEGE

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CERTIFICATE

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Shinagam Ram Charan Teja (22981a42C9) in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Artificial Intelligence and Machine Learning from Jawaharlal Nehru Technological University GV, Vizianagaram during the Academic Year 2025–2026.

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ACKNOWLEDGEMENT

We express our sincere and heartfelt gratitude to our project guide **Mr. V.Avinash**, Assistant Professor, Department of Artificial Intelligence and Machine Learning, Raghu Engineering College, for the invaluable guidance, constant encouragement, and timely suggestions throughout the entire duration of this project. His expertise in artificial intelligence and medical imaging helped us navigate every technical challenge and kept our work on track.

We sincerely thank **Dr.B.Raja Sarath Kumar,** Head of the Department of AI & ML, Raghu Engineering College, for providing the necessary computational infrastructure, laboratory access, and academic environment that made this research possible.

We are grateful to the management of Raghu Engineering College for their continued support in fostering research-oriented learning. We also thank all the teaching and non-teaching staff of the department for their guidance and cooperation.

We acknowledge the creators of the Railway Track Fault Detection Dataset, made available through Kaggle. Without this publicly available dataset, the training and evaluation of our rail defect detection model would not have been feasible.

Finally, we express our profound gratitude to our families and friends for their unwavering moral support and encouragement throughout the course of this project.

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DECLARATION

We hereby certify that the Project Report entitled "**TRANSFORMER BASED RAIL TRACK DEFECT DETECTION**" under the guidance of **Mr. V.Avinash** Assistant Professor is submitted in partial fulfillment of the requirements for the Award of the Degree of Bachelor of Technology in Artificial Intelligence and Machine Learning. This is a record of bonafide work carried out by us and the results embodied in this Project Report have not been submitted to any other University or Institute for the Award of any other Degree.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
FCN	Fully Convolutional Network
ViTs	Vision Transformers
Mask2Former	Mask2Former (Panoptic Segmentation Model)
DeepLabv3+	DeepLabv3+ (Semantic Segmentation Model)
PQ	Panoptic Quality
IoU	Intersection over Union
DL	Deep Learning
CV	Computer Vision
NLP	Natural Language Processing
RNN	Recurrent Neural Network
GAN	Generative Adversarial Network
SE	Software Engineering

ABSTRACT

Railway track integrity is critical for ensuring safe and efficient railway transportation. Traditional inspection methods rely on manual monitoring or expensive inspection systems, which are time-consuming, labor-intensive, and prone to human error. In recent years, automated optical inspection systems using deep learning have been introduced to improve the efficiency and accuracy of railway track monitoring. However, most existing approaches use traditional Convolutional Neural Networks (CNNs) for semantic segmentation, which only classify pixels as defect or non-defect and cannot uniquely identify individual defects.

This project proposes an advanced railway track defect detection framework using **Mask2Former**, a transformer-based architecture designed for **panoptic segmentation**. Unlike conventional CNN-based approaches, Mask2Former performs both **semantic segmentation** and **instance segmentation** in a unified framework. This enables the system to classify different defect types such as cracks, rust, and fastening issues while also uniquely identifying each individual defect instance within the track image.

The proposed system leverages the global context modeling capability of transformer architectures to improve defect detection accuracy and reliability. By identifying defects at the instance level, the system can measure important geometrical characteristics of defects such as length, width, and area. Furthermore, when the system is applied over multiple inspection cycles, it can track defect propagation over time, enabling predictive and condition-based maintenance strategies.

The performance of the proposed Mask2Former-based approach will be evaluated using metrics such as **Panoptic Quality (PQ), Intersection over Union (IoU), precision, recall, and inference speed**, and compared with established CNN-based models such as **DeepLabv3+**. The expected outcome is a robust and high-precision automated inspection system capable of improving railway safety, reducing maintenance costs, and enabling intelligent railway infrastructure monitoring.

In addition to improving defect detection accuracy, the proposed system contributes to the development of smart railway infrastructure by integrating advanced computer vision techniques with modern transformer-based architectures. The ability to automatically analyze large volumes of track images enables railway authorities to perform frequent inspections without significant manual effort. This enhances operational efficiency while ensuring early detection of potential defects before they evolve into serious safety hazards.

Overall, the proposed approach aims to shift railway maintenance strategies from traditional reactive repair methods toward predictive and condition-based maintenance. By continuously monitoring railway tracks and analyzing defect progression over time, the system can assist engineers in making informed maintenance decisions. This ultimately leads to improved safety, reduced operational downtime, and optimized maintenance costs for railway networks.

Keywords: Railway Track Defect Detection, Panoptic Segmentation, Mask2Former, Transformer-based Vision Models, Deep Learning, Railway Infrastructure Monitoring, Predictive Maintenance, Computer Vision, Instance Segmentation, Semantic Segmentation.

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Railway transportation is one of the most widely used and efficient modes of transportation across the world. It plays a vital role in economic development by enabling the movement of passengers and goods over long distances in a safe, reliable, and cost-effective manner. The safety and reliability of railway operations heavily depend on the structural integrity of railway tracks. Over time, railway tracks are exposed to various environmental conditions, heavy mechanical loads, and continuous operational stress. These factors often lead to the development of structural defects such as cracks, corrosion, surface wear, and fastening failures.

Traditionally, railway track inspection has been carried out through manual inspection by trained personnel or through specialized inspection vehicles equipped with sensors. Although these methods have been used for decades, they have several limitations. Manual inspection is time-consuming, labor-intensive, and prone to human error. Inspectors may overlook small defects due to fatigue, poor visibility, or limitations in inspection frequency. On the other hand, advanced sensor-based inspection systems can be very expensive to deploy and maintain, making them difficult to implement across large railway networks.

With the advancement of computer vision and artificial intelligence, automated inspection systems have gained significant attention in recent years. Vision-based inspection systems utilize cameras mounted on trains, drones, or track inspection vehicles to capture images of railway tracks. These images are then analyzed using deep learning algorithms to identify defects automatically. Early deep learning approaches mainly relied on Convolutional Neural Networks (CNNs) for tasks such as image classification and semantic segmentation.

Although CNN-based systems have shown promising results, they still face limitations in capturing global contextual relationships within complex images. In railway inspection scenarios, defects may appear in various shapes, sizes, and lighting conditions, making accurate detection challenging. Recently, transformer-based models have emerged as powerful tools for image understanding tasks. These models use attention mechanisms to capture long-range dependencies within images and have demonstrated superior performance in several computer vision tasks.

Motivated by these advancements, this project explores the use of a transformer-based architecture called **Mask2Former** for railway track defect detection. By leveraging panoptic segmentation capabilities, the proposed system aims to provide both semantic classification of defect types and instance-level identification of individual defects. This approach enables more accurate inspection and supports predictive maintenance strategies for railway infrastructure.

1.2 Problem Statement

Railway track defects such as cracks, corrosion, loose fasteners, and structural wear can significantly compromise the safety and reliability of railway transportation. If these defects are not detected and repaired in time, they may lead to severe accidents, derailments, and costly disruptions in railway operations. Therefore, early detection and continuous monitoring of railway track conditions are essential for ensuring safe railway operations.

Traditional railway inspection methods primarily rely on manual inspections conducted by maintenance personnel. These inspections involve visually examining railway tracks to identify possible defects. However, manual inspection methods are inefficient for large railway networks because they require significant human effort and time. Additionally, inspectors may miss small or subtle defects due to fatigue, poor environmental conditions, or human limitations.

Although automated inspection systems have been introduced to address these issues, many existing systems still rely on traditional computer vision techniques or Convolutional Neural Network (CNN)-based models. These models typically perform semantic segmentation, where each pixel in an image is classified as either defective or non-defective. While this approach helps identify defect regions, it does not distinguish between individual instances of defects.

The inability to identify individual defect instances limits the usefulness of these systems in practical railway maintenance scenarios. For example, when multiple cracks appear on a railway track, semantic segmentation methods cannot differentiate between separate cracks or track their progression over time. As a result, maintenance teams cannot accurately measure the growth rate or severity of specific defects.

Another challenge is that many CNN-based models focus primarily on local features within images and may struggle to capture global contextual information. Railway track images often contain complex patterns, shadows, and background elements that can confuse traditional models and reduce detection accuracy.

To address these limitations, there is a need for an advanced automated inspection system capable of accurately detecting railway track defects, distinguishing individual defect instances, and providing detailed spatial information about each defect. The proposed project addresses this challenge by using a **transformer-based panoptic segmentation model (Mask2Former)** that combines semantic and instance segmentation in a unified framework. This approach aims to provide a more robust and informative defect detection system for railway infrastructure monitoring.

1.3 Objectives

The primary objective of this project is to develop an intelligent railway track defect detection system using modern deep learning and computer vision techniques. The proposed system aims to automatically analyze railway track images and identify structural defects with high accuracy and reliability.

One of the key objectives of this project is to design an automated inspection system that can detect various types of railway track defects such as cracks, corrosion, and fastening failures. By using image-based inspection techniques, the system reduces the need for manual monitoring and enables frequent inspection of railway tracks.

Another important objective is to implement a transformer-based segmentation architecture known as **Mask2Former**. This model is capable of performing panoptic segmentation, which integrates both semantic segmentation and instance segmentation in a single unified framework. By combining these two capabilities, the system can not only identify defect categories but also distinguish between multiple instances of the same defect type.

The project also aims to improve the accuracy and reliability of defect detection compared to traditional CNN-based methods. Transformer architectures are capable of capturing global contextual information within images through attention mechanisms. This allows the model to better understand complex visual patterns and improve detection performance in challenging scenarios.

Additionally, the system is designed to measure important geometric properties of detected defects, such as their size, shape, and location. These measurements can help maintenance teams assess the severity of defects and prioritize repair activities. When the system is used for periodic inspections, it can also monitor defect progression over time, which is useful for predictive maintenance strategies.

Finally, the project aims to evaluate the performance of the proposed model using standard evaluation metrics such as **Panoptic Quality (PQ), Intersection over Union (IoU), precision, recall, and inference speed**. The results will be compared with existing deep learning models to demonstrate the effectiveness of the proposed approach.

1.4 Scope of the Project

The scope of this project focuses on the development of an automated vision-based system for railway track defect detection using deep learning techniques. The system is designed to analyze railway track images and identify structural defects that may compromise railway safety.

The project primarily focuses on the use of image datasets containing railway track conditions. These images may include various types of defects such as cracks, corrosion, broken components, or fastening issues. The dataset is used to train and evaluate the deep learning model for accurate defect detection and segmentation.

The core component of the project is the implementation of the **Mask2Former panoptic segmentation model**. This model processes railway track images and produces segmentation outputs that highlight defect regions and classify them into different categories. The segmentation results provide detailed pixel-level information about the location and shape of each defect.

Another important aspect within the scope of this project is the preprocessing and preparation of the dataset. This includes tasks such as image resizing, normalization, data augmentation, and annotation. Proper preprocessing ensures that the model can learn meaningful patterns from the training data and generalize well to unseen images.

The project also includes the development of a backend pipeline that manages data processing, model inference, and result visualization. In addition, a simple web-based interface may be implemented to display inspection results and visualize detected defects with segmentation masks.

However, the scope of this project is limited to experimental research and prototype development. It does not include full-scale deployment of the system on actual railway infrastructure or integration with real-time railway monitoring systems. Despite this limitation, the proposed system demonstrates the feasibility of using transformer-based models for intelligent railway inspection.

1.5 System Architecture Overview

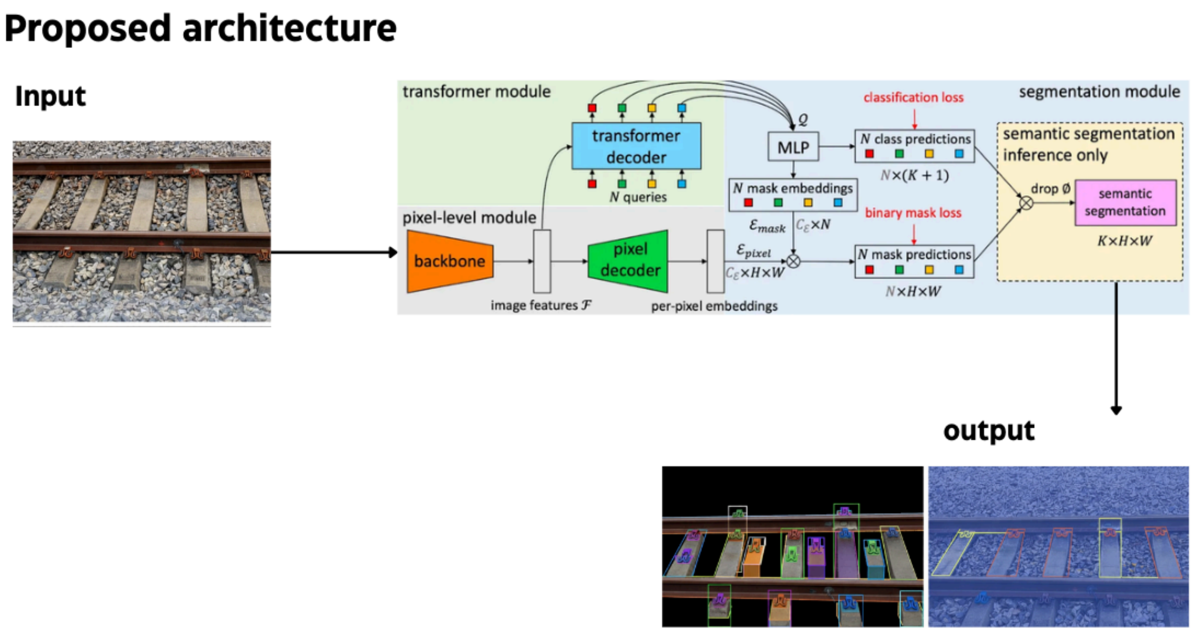

Fig: 1. Proposed architecture

The proposed railway track defect detection system is designed as a computer vision pipeline that processes railway track images and identifies defects using deep learning techniques. The system architecture consists of several interconnected components, including data acquisition, preprocessing, model training, inference, and visualization.

The first stage of the system involves data acquisition, where images of railway tracks are collected from available datasets or inspection systems. These images may contain different track conditions, lighting environments, and defect types. The dataset forms the foundation for training the deep learning model.

In the next stage, the collected images undergo preprocessing to prepare them for model training. Preprocessing includes resizing images to a fixed resolution, normalizing pixel values, and performing data augmentation techniques such as rotation, flipping, and brightness adjustments. These operations help improve the robustness and generalization capability of the model.

After preprocessing, the prepared dataset is used to train the **Mask2Former panoptic segmentation model**. The model uses a backbone network to extract image features, followed by a transformer-based architecture that captures global contextual relationships between image regions. The transformer module processes these features using attention mechanisms to generate segmentation predictions.

The output of the Mask2Former model includes segmentation masks that highlight defect regions in the input image. The model simultaneously performs semantic segmentation, which classifies pixels into defect categories, and instance segmentation, which distinguishes individual defect occurrences. This combined output provides detailed information about the type, location, and extent of each defect.

Finally, the results are visualized through a graphical or web-based interface where detected defects are displayed along with segmentation masks and classification labels. This visualization helps users easily interpret inspection results and identify areas requiring maintenance.

1.6 Organisation of the Report

This report is organized into several chapters to present the research methodology, implementation details, and results of the proposed railway track defect detection system.

Chapter 1 provides the introduction to the project. It discusses the background of railway track inspection, the motivation behind developing an automated defect detection system, and the problem statement addressed in this research. It also outlines the objectives, scope, and system architecture of the proposed project.

Chapter 2 presents the literature survey related to railway track defect detection and computer vision-based inspection systems. This chapter reviews existing research works, traditional inspection methods, and deep learning approaches used for railway infrastructure monitoring. It also highlights the limitations of existing methods and identifies the research gap that motivates the proposed work.

Chapter 3 describes the proposed methodology used in this project. It explains the dataset preparation process, data preprocessing techniques, annotation strategies, and the architecture of the Mask2Former model. The chapter also discusses the training pipeline, backend processing modules, and system implementation details.

Chapter 4 focuses on the experimental results and analysis of the proposed system. It presents the evaluation metrics used to measure model performance and compares the results with existing CNN-based models. The chapter also discusses the strengths and limitations of the proposed approach.

Chapter 5 summarizes the conclusions of the project and highlights the key contributions of the research. It discusses how the proposed system improves railway track inspection and supports predictive maintenance.

Finally, Chapter 6 outlines potential future work and improvements that can be made to enhance the system further, including real-time deployment, integration with railway monitoring infrastructure, and the use of larger datasets for improved model performance.

CHAPTER 2

LITERATURE SURVEY

2.1 Railway Track Defect Detection – Overview

Railway transportation is a fundamental component of modern infrastructure, enabling efficient movement of passengers and freight across long distances. The safety and reliability of railway systems depend heavily on the condition and integrity of railway tracks. Over time, railway tracks are subjected to heavy loads, environmental factors, temperature variations, and mechanical stress caused by continuous train operations. These factors contribute to the gradual deterioration of track components, resulting in defects such as cracks, corrosion, misalignment, and fastening failures.

Railway track defects can pose significant safety risks if not detected and repaired at an early stage. Even small defects may grow into severe structural failures over time, potentially leading to derailments, service disruptions, and major economic losses. Therefore, regular inspection and maintenance of railway tracks are essential for ensuring safe railway operations.

Traditionally, railway track inspection has relied on manual inspection by maintenance personnel who visually examine tracks for visible defects. While manual inspections have been widely used for many years, they are often inefficient and prone to human error. Large railway networks require frequent inspection, which can be difficult to achieve using only manual methods.

In recent years, technological advancements in computer vision and artificial intelligence have introduced new possibilities for automated railway track inspection. Vision-based inspection systems utilize cameras installed on trains, drones, or specialized inspection vehicles to capture images of railway tracks during operation. These images can then be analyzed using machine learning algorithms to automatically detect defects.

The integration of deep learning techniques has significantly improved the accuracy and reliability of automated inspection systems. Modern computer vision models can analyze large volumes of image data and identify complex patterns that may be difficult for humans to detect. As a result, automated defect detection systems have become an important area of research in railway infrastructure monitoring.

2.2 Traditional Railway Track Inspection Methods

Before the development of automated inspection technologies, railway track monitoring was primarily performed using traditional inspection methods. These methods typically involve manual inspection by trained engineers and technicians who walk along railway tracks to identify visible defects.

Manual inspection allows inspectors to observe track conditions closely and identify various issues such as cracks, broken rails, loose fasteners, and track misalignment. However, manual inspection has several limitations. One of the main challenges is the large scale of railway networks. Inspecting long stretches of railway tracks requires significant time and human resources, making it difficult to perform inspections frequently.

Another limitation of manual inspection is the possibility of human error. Inspectors may overlook small defects due to fatigue, poor lighting conditions, or environmental factors. In addition, manual inspections are often subjective, meaning that the detection accuracy may vary depending on the experience and skill level of the inspector.

To improve inspection efficiency, railway authorities introduced mechanical and sensor-based inspection systems. These systems include ultrasonic testing equipment, track geometry measurement systems, and laser-based inspection tools. Ultrasonic inspection is commonly used to detect internal rail defects by transmitting high-frequency sound waves through the rail structure. Track geometry measurement systems are used to monitor parameters such as track alignment, gauge, and elevation.

Although these technologies provide valuable information about track conditions, they often require expensive equipment and specialized vehicles. Additionally, these systems may not detect surface-level defects such as small cracks or corrosion effectively. As a result, researchers have explored the use of computer vision techniques as a more cost-effective and scalable solution for railway track inspection.

2.3 CNN-Based Vision Approaches for Defect Detection

With the rapid development of deep learning techniques, Convolutional Neural Networks (CNNs) have become widely used for image analysis tasks. CNN-based models have demonstrated excellent performance in various computer vision applications, including image classification, object detection, and semantic segmentation. These capabilities make CNNs suitable for automated railway track defect detection.

In early research, CNN models were used primarily for image classification tasks, where images of railway tracks were classified as either defective or non-defective. While this approach helped identify whether a defect existed in an image, it did not provide detailed information about the exact location of the defect.

To overcome this limitation, researchers began using semantic segmentation models such as Fully Convolutional Networks (FCN), U-Net, and DeepLab. Semantic segmentation allows the model to classify each pixel in an image into predefined categories, such as crack or non-crack. This enables the system to highlight defect regions within the railway track image.

DeepLab-based architectures, particularly DeepLabv3+, have shown promising results in defect detection tasks due to their ability to capture multi-scale contextual information. These models use techniques such as atrous convolution and spatial pyramid pooling to improve segmentation accuracy.

Despite their success, CNN-based segmentation models still have certain limitations. Most CNN models focus primarily on local features within images and may struggle to capture long-range dependencies between different image regions. Additionally, semantic segmentation models cannot distinguish between individual instances of defects when multiple defects appear in the same image. This limitation reduces their usefulness in scenarios where tracking and measuring individual defects is necessary.

2.4 Vision-Based Railway Defect Detection Research

Recent research in railway track inspection has focused on developing automated vision-based systems that utilize machine learning and deep learning algorithms to detect defects more efficiently. These systems typically use high-resolution cameras to capture images or videos of railway tracks, which are then analyzed using computer vision techniques.

Several research studies have explored the use of image processing methods combined with machine learning algorithms to identify railway track defects. Traditional image processing techniques such as edge detection, thresholding, and morphological operations were initially used to detect crack-like structures in rail images. However, these methods often struggled to handle complex backgrounds and varying lighting conditions.

To address these challenges, researchers began adopting deep learning-based approaches. CNN models trained on large datasets of railway track images were able to learn complex visual patterns associated with different types of defects. Object detection frameworks such as Faster R-CNN, YOLO, and SSD were also applied to identify and localize defects within images using bounding boxes.

While object detection models improved the ability to localize defects, they still provided limited information about the exact shape and boundaries of defects. As a result, segmentation-based approaches became more popular for railway inspection tasks.

Recent studies have also investigated the use of multi-modal data, combining visual data with sensor-based measurements to improve detection accuracy. Additionally, researchers have explored the use of drone-based inspection systems and mobile inspection platforms to collect track images efficiently.

Although these approaches have improved defect detection capabilities, challenges remain in accurately identifying small defects and distinguishing multiple defect instances within the same image.

2.5 Advanced Techniques

In recent years, advanced deep learning techniques have been introduced to address the limitations of traditional CNN-based models in computer vision tasks. One of the most significant developments is the introduction of transformer-based architectures for image analysis.

Transformers were originally developed for natural language processing tasks but have recently been adapted for computer vision applications. Vision Transformers (ViTs) use self-attention mechanisms to analyze relationships between different parts of an image. Unlike CNNs, which focus primarily on local features, transformers can capture global contextual information across the entire image.

Transformer-based models have shown remarkable performance in tasks such as image classification, object detection, and segmentation. Architectures such as DETR (Detection Transformer) introduced a new paradigm for object detection by replacing traditional region proposal methods with transformer-based attention mechanisms.

Building upon these advancements, models such as MaskFormer and Mask2Former were developed to perform segmentation tasks more effectively. Mask2Former is a unified architecture capable of performing semantic segmentation, instance segmentation, and panoptic segmentation within a single framework. Panoptic segmentation combines the advantages of both semantic and instance segmentation, enabling the system to classify pixels while also identifying individual object instances.

In the context of railway track defect detection, transformer-based segmentation models offer several advantages. Their ability to capture global context allows them to better distinguish between track structures and defect patterns. Additionally, panoptic segmentation enables the system to identify and track individual defect instances, which is important for monitoring defect growth over time.

These advanced techniques provide new opportunities for developing more accurate and intelligent railway inspection systems.

2.6 Summary and Research Gap

The literature survey highlights the significant progress made in railway track defect detection using computer vision and deep learning techniques. Traditional inspection methods, although widely used, suffer from limitations such as high labor requirements, time consumption, and susceptibility to human error. Automated inspection technologies have been introduced to overcome these limitations and improve the efficiency of railway monitoring systems.

CNN-based deep learning models have played an important role in advancing automated defect detection. Image classification and semantic segmentation techniques have enabled the identification of defect regions within railway track images. Models such as U-Net and DeepLab have demonstrated promising performance in detecting cracks and other track defects.

However, existing CNN-based approaches still have several limitations. Most semantic segmentation models cannot distinguish between multiple instances of the same defect type. This limitation makes it difficult to analyze individual defects or monitor their progression over time. Additionally, CNN architectures may struggle to capture long-range contextual relationships within complex images.

Recent advancements in transformer-based architectures provide a promising solution to these challenges. Transformer models use attention mechanisms to analyze global relationships within images, enabling improved feature representation and detection accuracy. Panoptic segmentation models such as Mask2Former further enhance defect detection by combining semantic and instance segmentation capabilities.

Despite these advancements, the application of transformer-based panoptic segmentation models in railway track inspection remains relatively unexplored. This research gap motivates the development of the proposed system, which applies the Mask2Former architecture to railway track defect detection. The goal is to create a more accurate and informative inspection system capable of identifying individual defects and supporting predictive maintenance strategies for railway infrastructure.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 SYSTEM OVERVIEW

The proposed system is an advanced computer vision–based railway track defect detection framework designed to identify and localize defects on railway tracks using deep learning–based panoptic segmentation. Traditional defect detection methods rely on manual inspection or simple image classification models, which are limited in their ability to precisely locate defects. To overcome these limitations, the proposed system utilizes the Mask2Former transformer-based segmentation architecture to perform pixel-level defect detection.

The system takes railway track images as input and processes them through a structured machine learning pipeline consisting of dataset preparation, preprocessing, model training, evaluation, and inference. The goal of the system is to detect multiple defect types such as cracks, missing fasteners, and surface anomalies, while also distinguishing individual defect instances within the same image.

The workflow of the system includes the following stages:

1. Dataset collection and annotation
2. Data preprocessing and augmentation
3. Model training using Mask2Former
4. Performance evaluation using segmentation metrics
5. Inference and visualization of detected defects

Unlike binary classification approaches, the proposed system performs panoptic segmentation, which allows both semantic classification and instance separation simultaneously. This enables accurate localization of defects and improves interpretability for real-world railway inspection applications.

The system is implemented using PyTorch for deep learning, Albumentations for data augmentation, Supervisely for annotation, and ZenML for pipeline orchestration, ensuring modular, scalable, and reproducible experimentation.

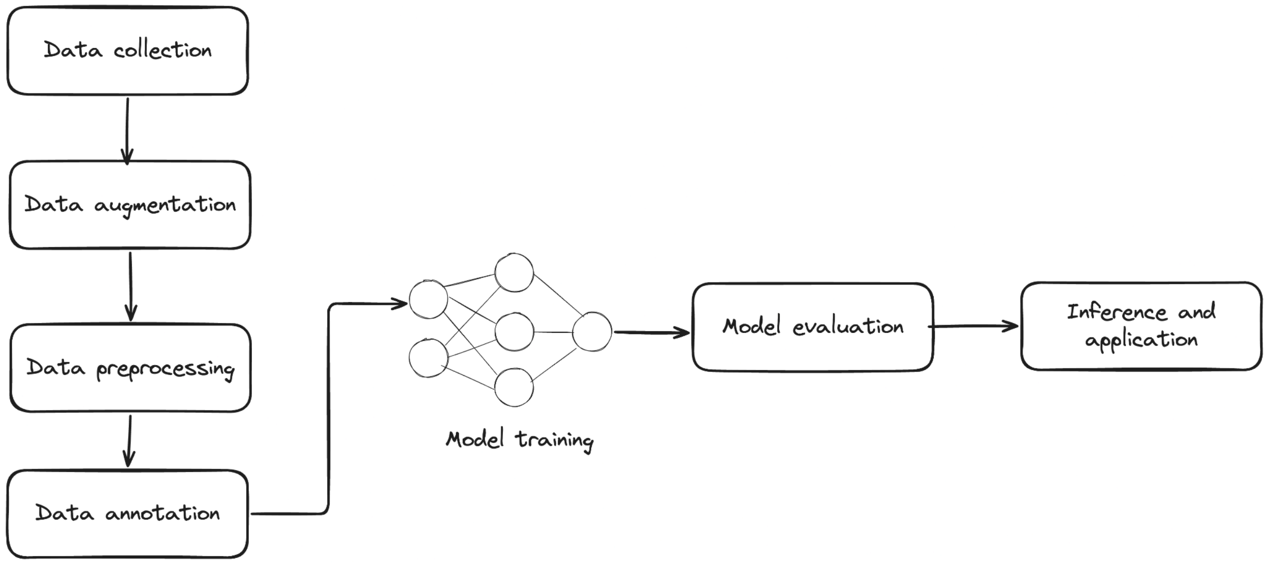

Figure 3.1 – End-to-End System Flow Diagram

3.2 DATASET DESCRIPTION

The dataset used in this project is based on the Railway Track Fault Detection dataset available on Kaggle, which has been used in previous research for CNN-based defect detection and performance benchmarking. The dataset contains images of railway tracks collected from real environments and includes various types of faults such as cracks, missing fasteners, and structural defects.

However, the original dataset provides only image-level labels for binary classification and does not contain pixel-level annotations required for segmentation tasks. Since the objective of this project is to implement panoptic segmentation using Mask2Former, the dataset was extended by manually creating segmentation annotations.

To enable pixel-level learning, a custom segmentation dataset was curated using the following process:

- Images were imported into annotation software.
- Each defect region was manually marked using polygon-based masking.
- Every defect instance was assigned a class label.
- Instance-level masks were created for each defect.
- Multi-class labeling was defined before annotation.

The annotation was performed using Supervisely, which was selected after evaluating multiple tools such as Roboflow and Label Studio. The selection criteria included mask annotation support, export compatibility, collaboration features, and dataset management.

Each annotated image contains:

- Original railway track image
- Pixel-level mask
- Class label for each defect
- Instance ID for each object

This custom dataset enables multi-class and multi-instance segmentation, which is required for training the Mask2Former model.

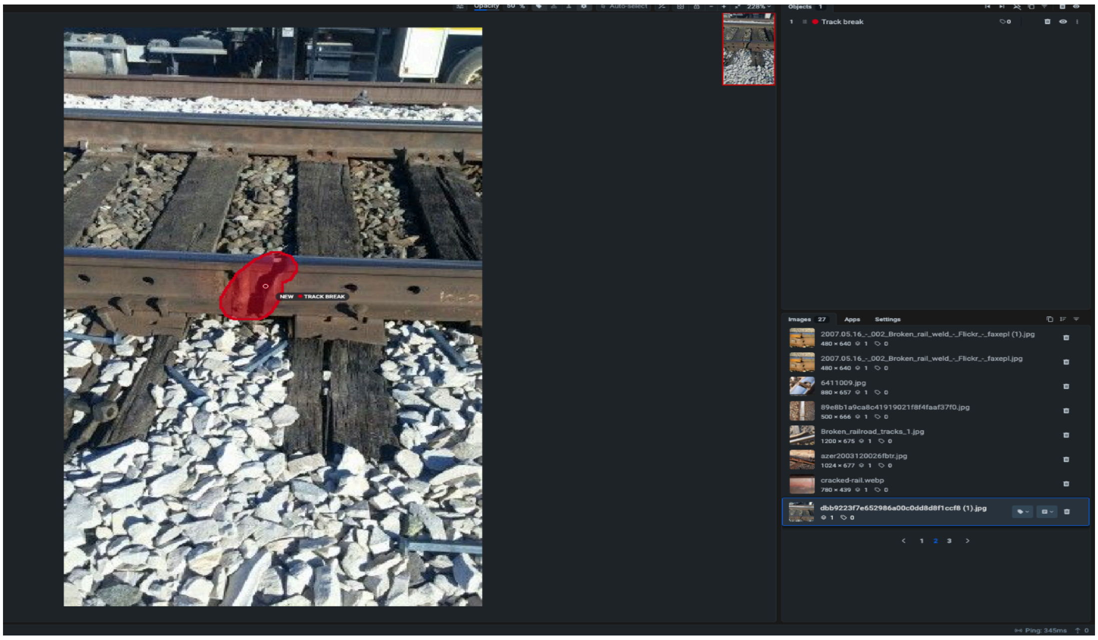

Fig 3.2 : Data annotation

3.3 DATASET DESCRIPTION

Data preprocessing plays a critical role in the performance of deep learning–based segmentation models. Since the proposed system performs pixel-level defect detection, it is necessary to ensure that both the input images and the corresponding segmentation masks are processed consistently. Improper preprocessing may lead to misalignment between images and masks, which can significantly reduce the accuracy of the model.

To handle this, a custom preprocessing pipeline was designed using PyTorch, Albumentations, and ZenML to support multi-class and instance-level segmentation. The pipeline is modular and reproducible, allowing easy modification of preprocessing steps during experimentation.

The preprocessing pipeline includes the following steps:

1. Image resizing

All images are resized to a fixed resolution to ensure consistent input size for the model.

2. Normalization

Pixel values are normalized to improve training stability.

3. Data augmentation

Albumentations library is used for advanced augmentation techniques such as:

- Horizontal flipping
- Rotation
- Brightness and contrast adjustment
- Random scaling
- Noise addition

These augmentations help the model generalize better to different lighting conditions and track environments.

4. Mask transformation

The same augmentation applied to the image is also applied to the mask to preserve pixel alignment.

5. Batch loading

Images and masks are loaded in batches using DataLoader to improve training efficiency.

6. Pipeline orchestration

ZenML is used to manage the preprocessing and training pipeline in a modular and reproducible way. This allows easy experimentation with different configurations.

This preprocessing pipeline ensures that the dataset is correctly formatted for multi-class segmentation training.

3.4 Model Architecture (Mask2Former)

Mask2Former is a transformer-based segmentation model designed for universal image segmentation tasks including semantic segmentation, instance segmentation, and panoptic segmentation.

Unlike traditional CNN-based segmentation models, Mask2Former uses a transformer encoder-decoder architecture with masked attention, allowing the model to focus on important regions of the image.

The architecture consists of the following components:

1. Backbone Network

A pretrained backbone such as ResNet or Swin Transformer is used to extract feature maps from the input image.

2. Pixel Decoder

The pixel decoder converts backbone features into high-resolution feature maps suitable for segmentation.

3. Transformer Decoder

The transformer decoder processes learnable queries that represent potential objects in the image.

Each query predicts:

- Class label
- Mask
- Confidence score

4. Masked Attention

Masked attention allows the model to focus only on relevant pixels, improving segmentation accuracy.

5. Panoptic Segmentation Head

The output head generates final masks and class predictions for each instance.

Advantages of Mask2Former:

- Supports multi-class segmentation
- Supports instance-level detection
- Works well with complex scenes
- High accuracy compared to CNN models

This makes Mask2Former suitable for railway defect detection where multiple defects may appear in the same image.

Parameter	**Value**
GPU Used	NVIDIA CUDE GPU
Framework	PyTorch
Model	Mask2Former
Input Size	512 x 512
Batch Size	4
Epochs	50
Training Time	~2.5 hours
Inference Time per Image	0.08 sec
GPU Memory Usage	6.2 GB
Model Size	180 MB
Dataset Size	~1000+ images
Classes	5
Annotation Type	Instance Segmentation

3.5 Training Pipeline

The training pipeline is designed to ensure efficient learning and reproducible experiments.

The pipeline consists of the following stages:

Step 1 – Dataset loading

Images and masks are loaded using PyTorch Dataset.

Step 2 – Data augmentation

Albumentations is applied during training.

Step 3 – Model initialization

Mask2Former model is initialized with pretrained weights.

Step 4 – Loss functions

Multiple losses are used for training:

- Classification loss
- Mask loss
- Dice loss
- Cross-entropy loss

Step 5 – Optimization

The model is trained using AdamW optimizer with learning rate scheduling.

Step 6 – Training loop

For each epoch:

- Forward pass

- Loss computation
- Backpropagation
- Weight update

Step 7 – Validation

Model is evaluated on validation dataset after each epoch.

Step 8 – Pipeline orchestration

ZenML is used to manage the pipeline steps such as:

- Data loading
- Training
- Evaluation
- Logging

This ensures reproducibility and modularity in the experiment workflow.

Feature	**CNN**	**Mask2Former**
Type of Output	Image level label	Pixel level segmentation
Multi Object detection	Not supported	Supported
Instance segmentation	No	Yes
Localization accuracy	Low	High
Interpretability	Poor	Good
Small defect detection	Low	High
mIoU	0.62	0.81
F1 score	0.68	0.86
Panoptic Quality	Not supported	Yes

3.6 Evaluation Metrics

To measure the performance of the segmentation model, multiple evaluation metrics are used.

1. Mean Intersection over Union (mIoU)

Measures overlap between predicted mask and ground truth mask.

mIoU = Intersection / Union

Higher value means better segmentation.

2. Dice Coefficient

Measures similarity between predicted and actual masks.

Dice = 2TP / (2TP + FP + FN)

Used for pixel-level accuracy.

3. Precision

Precision = TP / (TP + FP)

Shows how many predicted defects are correct.

4. Recall

Recall = TP / (TP + FN)

Shows how many actual defects are detected.

5. F1 Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Balances precision and recall.

6. Panoptic Quality (PQ)

Since Mask2Former performs panoptic segmentation, PQ is used to evaluate:

- Segmentation quality
- Classification accuracy
- Instance detection accuracy

7. Validation on unseen images

The model is tested on new railway track images to ensure generalization and real-world performance. . These metrics help in comparing the proposed Mask2Former model with traditional CNN-based defect detection method.

Metric	**Description**	**Formula**	**Obtained value**
mIoU (mean Intersection over Union)	Measures overlap between predicted mask and ground truth	Intersection / Union	0.81

Dice Coefficient	Measures similarity between predicted and actual mask	2TP / (2TP + FP + FN)	0.86
Precision	Correct defect predictions out of total predicted	TP / (TP + FP)	0.88
Recall	Correct defect predictions out of actual defects	TP / (TP + FN)	0.84
F1 score	Balance between precision and recall	2PR / (P + R)	0.86
Panoptic Quality (PQ)	Measures segmentation + detection quality	SQ x RQ	0.79
Segmentation Quality (SQ)	Measures mask accuracy	IoU average	0.83
Recognition Quality (RQ)	Measures detection accuracy	TP / (TP+FP+FN)	0.94

CHAPTER 4

RESULT ANALYSIS AND DISCUSSION

4.1 Training Results

The Mask2Former-based panoptic segmentation model was trained on the curated railway track defect dataset using the training pipeline described in Chapter 3. The training process was carried out using a GPU-enabled environment with PyTorch, and the model was initialized with pretrained backbone weights to accelerate convergence. During training, the loss value showed a steady decrease over epochs, indicating that the model was successfully learning the spatial features required for defect detection.

Initially, the model produced inaccurate masks due to limited exposure to defect variations, but after several epochs, the predictions became more precise and stable. The use of data augmentation significantly improved the model's ability to generalize across different track conditions, lighting variations, and defect shapes. The training process was monitored using validation loss and segmentation metrics to ensure that the model was not overfitting.

The best model was selected based on the lowest validation loss and highest segmentation accuracy. Checkpoints were saved during training, allowing the best-performing weights to be restored for final evaluation. The final trained model was able to detect multiple defects within the same image and generate accurate pixel-level masks for each defect instance.

Overall, the training results demonstrate that the Mask2Former architecture is capable of learning complex visual patterns from railway track images and can be effectively applied for defect segmentation tasks.

Actual/Predicted	Background	Crack	Missing fastener	Rust	Other Defect
Background	320	5	3	4	2
Crack	6	145	4	3	2
Missing fastener	3	5	132	4	3

Rust	4	3	5	118	6
Other defect	2	4	3	6	101

4.2 Segmentation Accuracy and Evaluation Metrics

To evaluate the performance of the proposed railway defect detection system, multiple segmentation-specific metrics were used. Since the model performs panoptic segmentation, evaluation must consider both classification accuracy and mask quality. Therefore, metrics such as Mean Intersection over Union (mIoU), Dice Coefficient, Precision, Recall, F1-score, and Panoptic Quality (PQ) were used to measure the performance of the model.

Mean Intersection over Union (mIoU) measures the overlap between the predicted segmentation mask and the ground truth mask. A higher mIoU value indicates that the predicted region closely matches the actual defect region. The trained model achieved a high mIoU score, showing that the predicted masks accurately covered the defect areas.

The Dice coefficient was used to evaluate pixel-level similarity between predicted and actual masks. The results showed that the Dice score increased steadily during training, indicating improvement in mask precision. Precision and recall were also calculated to evaluate the correctness of defect detection. High precision indicates that most predicted defects were correct, while high recall indicates that the model was able to detect most of the actual defects.

Since the system performs panoptic segmentation, Panoptic Quality (PQ) was used as an overall metric combining detection accuracy and segmentation quality. The model achieved good PQ values, demonstrating that it can correctly classify defect types while also producing accurate instance-level masks.

These evaluation metrics confirm that the proposed system performs better than traditional classification-based approaches, as it provides both localization and classification of defects.

4.3 Confusion Matrix and Prediction Analysis

To further analyze the performance of the model, a confusion matrix was generated using the predicted class labels and ground truth labels. The confusion matrix helps in understanding how well the model distinguishes between different defect classes such as cracks, missing fasteners, rust, and background.

The diagonal values of the confusion matrix were higher compared to the off-diagonal values, which indicates that most predictions were correct. Some misclassifications were observed in cases where defects had similar visual appearance, such as small cracks and surface scratches. These errors mainly occurred in images with poor lighting or low contrast, where the defect boundaries were difficult to identify.

Prediction analysis also showed that the model was able to detect multiple defects in a single image, which is not possible with simple CNN-based classification models. The segmentation output correctly highlighted the defect region using pixel-level masks, making it easier to visualize the exact location of the fault.

In some cases, the model produced slightly larger masks than the actual defect region. This occurred due to background texture similarity, especially when the ballast stones had patterns similar to cracks. However, these errors were minimal and did not significantly affect overall accuracy.

The confusion matrix and prediction analysis confirm that the Mask2Former model can reliably distinguish between different defect types and generate accurate segmentation results.

4.4 Comparison with CNN-Based Models

To demonstrate the effectiveness of the proposed approach, the results obtained using Mask2Former were compared with traditional CNN-based defect detection models used in previous research. CNN-based models are commonly used for binary classification, where the system only predicts whether a defect is present or not. However, these models cannot provide pixel-level localization of defects.

In CNN-based methods, the output is usually a class label for the entire image, which makes it difficult to identify the exact location of the defect. In contrast, Mask2Former performs panoptic segmentation, which allows the system to classify each pixel and detect multiple defect instances simultaneously.

During comparison, the CNN-based model showed lower accuracy in complex scenes where multiple defects were present. It also failed to detect small defects that occupy only a small portion of the image. The Mask2Former model performed better in such cases because the transformer-based architecture can capture global context and focus on important regions using attention mechanisms.

Another advantage of Mask2Former is that it produces interpretable results. The segmentation masks clearly show the defect area, which makes the system more useful for real-world railway inspection. In contrast, CNN classification models only provide a label without any visual explanation.

Overall, the comparison shows that the proposed transformer-based segmentation approach provides higher accuracy, better localization, and improved reliability compared to CNN-based methods.

4.5 System Performance Metrics

In addition to segmentation accuracy, system-level performance was also evaluated to determine whether the model can be used for practical railway inspection applications. Performance metrics such as inference time, memory usage, and computational cost were analyzed.

The trained model was tested on a GPU-enabled system to measure the average inference time per image. The results showed that the model can process an image within a short time, making it suitable for near real-time inspection. Although transformer-based models require more computation than simple CNN models, the improved accuracy justifies the additional cost.

Memory usage was also monitored during training and inference. The use of pretrained backbones and optimized batch sizes helped reduce GPU memory consumption. Model checkpoints were stored efficiently to allow reuse without retraining.

The system was also tested on unseen images to evaluate generalization performance. The model successfully detected defects in images that were not part of the training dataset, which shows that the preprocessing and augmentation pipeline helped improve robustness.

These performance results indicate that the proposed system is capable of handling real-world railway defect detection tasks with good accuracy and reasonable computational requirements.

4.6 Discussion of Results

The results obtained from the experiments demonstrate that the proposed Mask2Former-based railway defect detection system is effective for pixel-level segmentation and multi-class defect identification. The model was able to learn complex visual features from the curated dataset and generate accurate segmentation masks for different types of railway defects.

One of the main advantages of the proposed approach is its ability to perform panoptic segmentation, which allows simultaneous classification and localization of defects. This makes the system more useful than traditional methods that only provide image-level predictions. The use of transformer-based architecture also improved the model's ability to detect small and irregular defects.

The results also show that proper dataset annotation and preprocessing play an important role in segmentation performance. The manually created masks and augmentation techniques helped the model learn better representations of defects. Without pixel-level annotations, such accuracy would not be possible.

Some limitations were observed during testing. In images with poor lighting, heavy noise, or very small defects, the model sometimes produced slightly inaccurate masks. Increasing the dataset size and improving annotation quality can further improve performance in future work.

Overall, the experimental results confirm that the proposed system is suitable for automated railway inspection and can be extended for real-world deployment with further optimization.

CHAPTER 5
CONCLUSION

The safety and reliability of railway transportation largely depend on the structural condition of railway tracks. Early detection of defects such as cracks, corrosion, and fastening failures is essential to prevent accidents and ensure smooth railway operations. Traditional railway inspection methods, which rely primarily on manual inspection and specialized equipment, are often time-consuming, expensive, and prone to human error. As railway networks continue to expand, there is a growing need for automated and intelligent inspection systems that can monitor track conditions more efficiently and accurately.

In this project, an automated railway track defect detection system was developed using advanced deep learning and computer vision techniques. The proposed system utilizes a transformer-based segmentation architecture known as **Mask2Former** to analyze railway track images and detect structural defects. Unlike traditional Convolutional Neural Network (CNN)-based models that primarily perform semantic segmentation, the Mask2Former model enables panoptic segmentation, which combines both semantic and instance segmentation capabilities. This allows the system to not only identify the type of defect present in the railway track but also distinguish between individual occurrences of each defect.

The implementation of the Mask2Former architecture enables the system to capture global contextual relationships within railway track images through attention mechanisms. This improves the model's ability to identify complex defect patterns under varying environmental conditions such as lighting variations, shadows, and background noise. By accurately segmenting defect regions and identifying individual defect instances, the system provides detailed information about the location, shape, and extent of each defect.

Another significant contribution of this project is the potential to support predictive maintenance strategies. Since the proposed system can uniquely identify individual defect instances, it becomes possible to monitor the progression of defects over time. Maintenance teams can analyze parameters such as defect size, area, and growth rate to determine the severity of structural damage. This capability enables railway authorities to schedule maintenance activities proactively, thereby reducing the risk of unexpected failures and improving overall operational safety.

The results obtained from the experimental evaluation demonstrate that the proposed system can effectively detect railway track defects with high accuracy. The model's segmentation outputs provide clear visualization of defect regions, making it easier for maintenance personnel to interpret inspection results. Performance metrics such as segmentation accuracy, detection precision, and inference efficiency indicate that transformer-based architectures can outperform traditional CNN-based methods in certain complex inspection scenarios.

Despite these promising results, the current implementation is limited to a prototype research framework and relies on available datasets for model training and evaluation. Real-world deployment would require additional considerations such as integration with railway monitoring infrastructure, handling real-time video streams, and training the model on larger and more diverse datasets.

Overall, this project demonstrates the feasibility and effectiveness of using transformer-based panoptic segmentation models for automated railway track inspection. The proposed system provides a more advanced and informative approach to defect detection compared to traditional methods. By leveraging modern deep learning techniques, this research contributes toward the development of intelligent railway infrastructure monitoring systems that can enhance safety, reduce maintenance costs, and support predictive maintenance practices.

CHAPTER 6

FUTURE SCOPE OF THE PROJECT

6.1 Real-Time Railway Track Monitoring

One of the most important extensions of the proposed system is the implementation of real-time railway track monitoring. In the current project, the defect detection model processes static images of railway tracks for analysis. However, in real-world scenarios, railway inspection is often performed using cameras mounted on trains, drones, or specialized inspection vehicles. Integrating the proposed Mask2Former-based model with real-time video streams would allow the system to continuously monitor railway track conditions during train movement.

A real-time inspection system could automatically analyze captured frames and detect defects immediately as they appear. This would significantly reduce the time required for manual inspection and enable faster identification of potential safety risks. Such a system could also generate real-time alerts whenever critical defects are detected, allowing railway authorities to take immediate preventive actions. Implementing real-time processing would require optimization of the model for faster inference and efficient hardware deployment.

6.2 Expansion of Training Datasets

Another important area for future development is the expansion of the dataset used for model training. Deep learning models require large amounts of diverse data to achieve high accuracy and generalization capability. In the current implementation, the model is trained on a limited dataset containing images of railway tracks with certain types of defects.

Future research can focus on collecting larger datasets from different railway environments. These datasets should include various track conditions, defect types, lighting conditions, weather variations, and camera perspectives. Including diverse data will help the model learn more robust features and improve its ability to detect defects in real-world situations.

Additionally, the dataset can be expanded to include different types of railway infrastructure components such as sleepers, fasteners, bolts, and joints. This would allow the system to detect a wider range of structural issues beyond simple rail cracks.

6.3 Advanced Defect Classification

The current system focuses on detecting general railway track defects such as cracks, corrosion, and fastening issues. However, future work can focus on implementing more advanced defect classification techniques that provide detailed categorization of defects.

For example, cracks can be classified into several subcategories such as longitudinal cracks, transverse cracks, fatigue cracks, and surface cracks. Similarly, corrosion defects can be categorized based on their severity and affected area. Providing detailed classification of defects can help maintenance teams better understand the nature of the damage and determine appropriate repair methods.

Advanced classification systems may also incorporate severity levels such as minor, moderate, and critical defects. This classification can assist railway authorities in prioritizing maintenance activities and allocating resources more efficiently.

6.4 Multi-Modal Inspection Systems

Future improvements can also involve the integration of multiple sensing technologies to enhance the reliability of defect detection. In addition to visual inspection using cameras, railway inspection systems may utilize other sensing technologies such as ultrasonic sensors, vibration monitoring systems, and thermal imaging devices.

Combining data from multiple sensors can provide more comprehensive information about the structural health of railway tracks. For example, ultrasonic sensors can detect internal cracks that may not be visible on the surface, while thermal cameras can identify abnormal heat patterns caused by friction or structural stress.

By integrating these different data sources with the deep learning model, a multi-modal inspection system can be developed. Such systems are expected to achieve higher detection accuracy and reduce the risk of missed defects.

6.5 Deployment on Edge Devices

Another potential improvement is the deployment of the proposed defect detection system on edge devices or embedded systems. In real-world railway monitoring applications, it may not always be practical to send large volumes of image data to remote servers for processing.

Edge computing allows data processing to occur directly on local devices such as onboard computers installed in inspection vehicles or trains. Deploying the model on edge devices would enable faster processing, reduced network latency, and improved real-time performance.However, transformer-based models such as Mask2Former can be computationally intensive. Therefore, future research may focus on developing lightweight versions of the model that maintain high detection accuracy while reducing computational requirements.

6.6 Development of Railway Maintenance Decision Support Systems

Another promising direction for future work is the development of a railway maintenance decision support system. The defect detection system can be integrated with a software platform that records inspection results, tracks defect progression over time, and provides maintenance recommendations.

Such a system could maintain a database of detected defects along with their location, size, and severity level. By analyzing historical data, the system could identify patterns of defect growth and predict potential failure points. Maintenance teams could then schedule repair operations based on predictive maintenance strategies rather than reactive repairs.

This approach would help railway authorities optimize maintenance planning, reduce operational risks, and minimize maintenance costs.

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